

Colligative properties

1. Define or explain the following terms :

(a) Colligative properties (b) Raoult's law (c) Boiling point elevation (d) Boiling point constant (e) Freezing point depression

(a) Colligative Properties (Greek *colligatus* = Collected together).

A colligative property may be defined as one which depends on the number of particles in solution and not in any way on the size or chemical nature of the particles.

Dilute solutions containing non-volatile solute exhibit the following properties :

- (1) Lowering of the Vapour Pressure
- (2) Elevation of the Boiling Point
- (3) Depression of the Freezing Point
- (4) Osmotic Pressure

The colligative properties of dilute solutions are particularly important as these provide valuable methods for finding the molecular weights of the dissolved substances.

Raoult's law

The relative lowering of the vapour pressure of a dilute solution is equal to the mole fraction of the solute present in dilute solution.

The vapour pressure of a pure solvent is decreased when a non-volatile solute is dissolved in it.

If p is the vapour pressure of the solvent and p_s that of the solution, the lowering of vapour pressure is $(p - p_s)$.

$$\text{Relative Lowering of Vapour Pressure} = \frac{p - p_s}{p}$$

Raoult's Law can be expressed mathematically in the form: $\frac{p - p_s}{p} = \frac{n}{n + N}$

where n = number of moles or molecules of solute

N = number of moles or molecules of solvent.

2. Derive Roults Law from lowering of vapour pressure.

The vapour pressure of the pure solvent is caused by the number of molecules evaporating from its surface. When a nonvolatile solute is dissolved in solution, the presence of solute molecules in the surface blocks a fraction of the surface where no evaporation can take place.

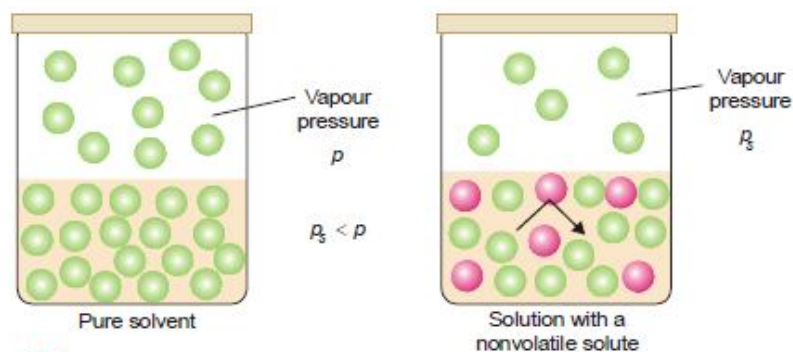


Figure: Lowering of vapour pressure by a nonvolatile solute. The particles of the solute block the escape of solvent molecules from the surface of the solution.

This causes the lowering of the vapour pressure. The vapour pressure of the solution is, therefore, determined by the number of molecules of the solvent present at any time in the surface which is proportional to the mole fraction.

That is, $P_s \propto \frac{N}{n+N}$ where N = moles of solvent and n = moles of solute.

Or $P_s = k \frac{N}{n+N}$ (1) k is proportionally constant

In case of pure solvent $n = 0$ and hence mole fraction of solvent = $\frac{N}{n+N} = \frac{N}{0+N} = 1$

Now from eq. 1 the vapour pressure $p = k$

Therefore the eq. 1 assumes the form $P_s = P \frac{N}{n+N}$

$$\frac{P_s}{P} = \frac{N}{n+N}$$

$$\text{Or, } 1 - \frac{P_s}{P} = 1 - \frac{N}{n+N}$$

$$\text{Or, } \frac{P - P_s}{P} = \frac{n}{n+N}, \text{ this is Raoult's law.}$$

Problem : Calculate the vapour pressure lowering caused by the addition of 100 g of sucrose (mol mass = 342) to 1000 g of water if the vapour pressure of pure water at 25°C is 23.8 mm Hg.

Using Raoult's Law Equation

$$\frac{P - P_s}{P} = \frac{\Delta p}{P} = \frac{n}{n+N} \quad \dots(1)$$

where

Δp = lowering of vapour pressure

P = vapour pressure of water = 23.8 mm Hg

n = moles of sucrose = $\frac{100}{342} = 0.292$ mole

N = moles of water = $\frac{1000}{18} = 55.5$ moles

Substituting values in equation (1)

$$\frac{\Delta p}{23.8} = \frac{0.292}{0.292 + 55.5}$$

$$\Delta p = 23.8 \times \frac{0.292}{55.792} = 0.125 \text{ mm}$$

Thus the lowering of vapour pressure = 0.125 mm Hg

Elevation of boiling point:

When a liquid is heated, its vapour pressure rises and when it equals the atmospheric pressure, the liquid boils. The addition of a non volatile solute lowers the vapour pressure and consequently elevates the boiling point as the solution has to be heated to a higher temperature to make its vapour pressure become equal to atmospheric pressure. If T_b is the boiling point of the solvent and T is the boiling point of the solution, the difference in the boiling points (ΔT) is called the **elevation of boiling point**. $T - T_b = \Delta T$

Relation between Boiling point elevation and vapour pressure lowering

The vapour pressure curves of the pure solvent, and solutions (1) and (2) with different concentrations of solute are shown in Fig.2

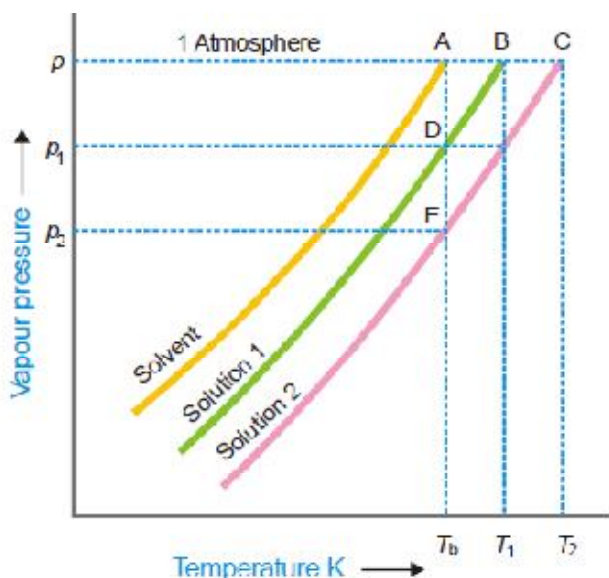


Fig.2: Ostwald-Walker method of measuring the relative lowering of vapour pressure.

For dilute solutions, the curves BD and CE are parallel and straight lines approximately. Therefore

for similar triangles ACE and ABD , we have $\frac{AB}{AC} = \frac{AD}{AE}$ or $\frac{T_1 - T_b}{T_2 - T_b} = \frac{p - p_1}{p - p_2}$

where $p - p_1$ and $p - p_2$ are lowering of vapour pressure for solution 1 and solution 2 respectively. Hence the elevation of boiling point is directly proportional to the lowering of vapour pressure.

Or, $\Delta T \propto P - P_s$

Freezing point depression /relation between freezing point depression and vapour-pressure lowering

The vapour pressure of a pure liquid changes with temperature as shown by the curve ABC , in Fig. 3. There is a sharp break at B where, in fact, the freezing-point curve commences. Thus the point B corresponds to the freezing point of pure solvent, T_f . The vapour pressure curve of a solution (*solution 1*) of a nonvolatile solute in the same solvent is also shown in Fig.3 It is similar to the vapour pressure curve of the pure solvent and meets the freezing point curve at F , indicating that T_1 is the freezing point of the solution. The difference of the freezing point of the pure solvent and the solution is referred to as the Depression of freezing point. It is represented by the symbol ΔT or ΔT_f .

$$T_f - T_1 = \Delta T$$

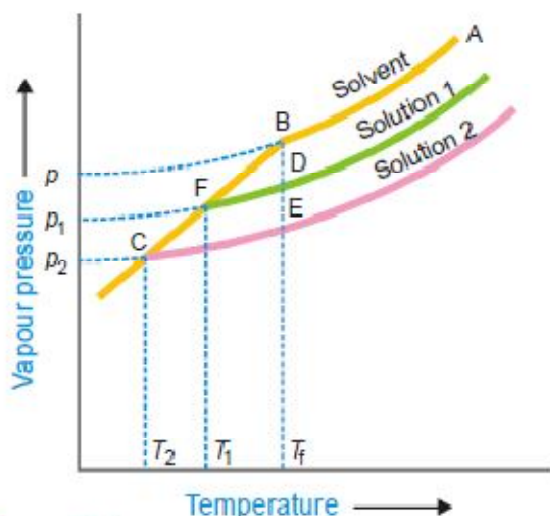


Fig. 3: Relation between lowering of vapour-pressure and depression of freezing point.

When more of the solute is added to the solution 1, we get a more concentrated solution (*solution 2.*) The vapour pressure of *solution 2* meets the freezing-point at C , indicating a further lowering of freezing point to T_2 . For dilute solutions FD and CE are approximately parallel straight lines and BC is also a straight line. Since the triangles BDF and BEC are similar, $\frac{DF}{EC} = \frac{BD}{BE}$ or, $\frac{T_f - T_1}{T_f - T_2} = \frac{p - p_1}{p - p_2}$ where p_1 and p_2 are vapour pressure of *solution 1* and *solution 2* respectively. Hence depression of freezing point is directly proportional to the lowering of vapour pressure. Or $\Delta T \propto P - P_s$